

IBM University Engagement Project Submission

“Exploring the Power of Agentic AI with IBM Granite and LangFlow”

Project Tile – **AI-Powered Investment Analyst**

Domain of Project – **Finance / FinTech**

Student Name	Recent Passport Photo	Email ID	Mobile number [WhatsApp]
Sujal Bhagat		sujalbhagatcodes@gmail.com	8329014204

Recommended font style and size:

Font Style: Arial

Font Size (Heading): 28

Font Size (Content): 20

Problem Statement -

Problem Statement No. 4 – Sequential Task Agent: Investment Analyst

The Challenge

Investment analysts rely on multiple sources of financial information such as company financial statements, stock market performance, quarterly earnings, analyst reports, economic indicators, and financial news. Since this information is distributed across different platforms, analysts spend significant time collecting, validating, and interpreting the data before making investment decisions.

Manual investment analysis is often slow, repetitive, and susceptible to human error. Furthermore, rapidly changing market conditions require continuous monitoring, making it difficult for analysts and investors to respond quickly to new opportunities or risks.

There is a growing need for an intelligent AI system capable of automating sequential financial analysis while producing reliable and data-driven investment recommendations.

Objectives

- Automatically collect financial data from trusted sources.
- Calculate financial ratios.
- Compare companies with industry competitors..
- Detect investment risks

Proposed Solution -

Proposed Solution

The proposed solution is an AI-powered Investment Analyst developed using IBM Orchestrate, IBM Granite Models, HTML, CSS, JavaScript, and financial APIs.

The system automates the complete investment analysis workflow through a Sequential Task Agent.

Instead of manually collecting financial information, the agent performs every task automatically in a predefined sequence.

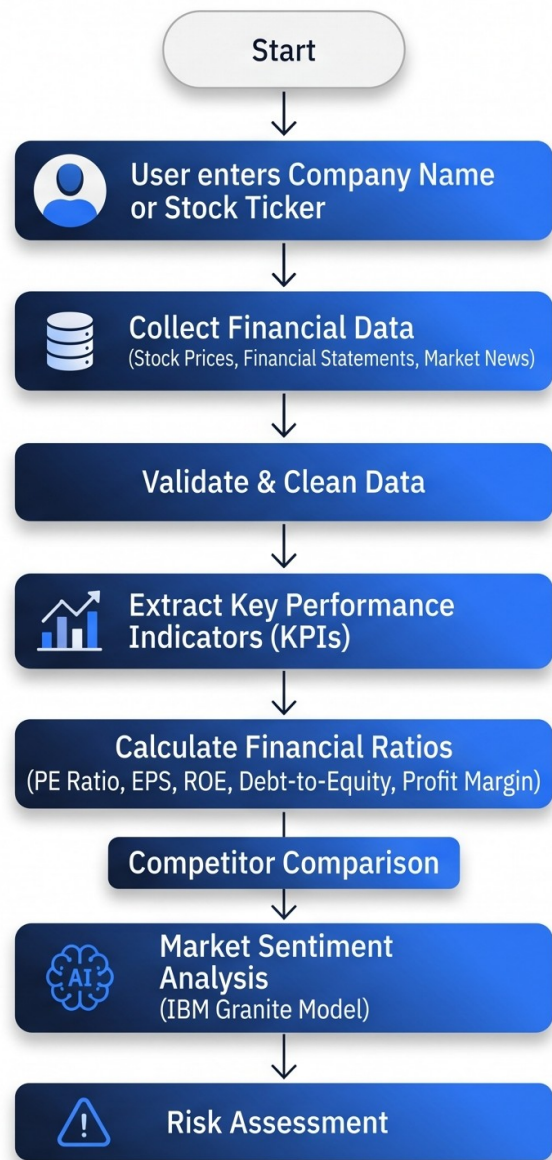
Sequential Workflow:

User Request → Collect Financial Data → Validate Information → Extract KPIs → Calculate Financial Ratios → Compare Competitors → Analyze Market Sentiment → Assess Investment Risk → Generate AI Recommendation → Display Dashboard → Generate Investment Report

Technology Used

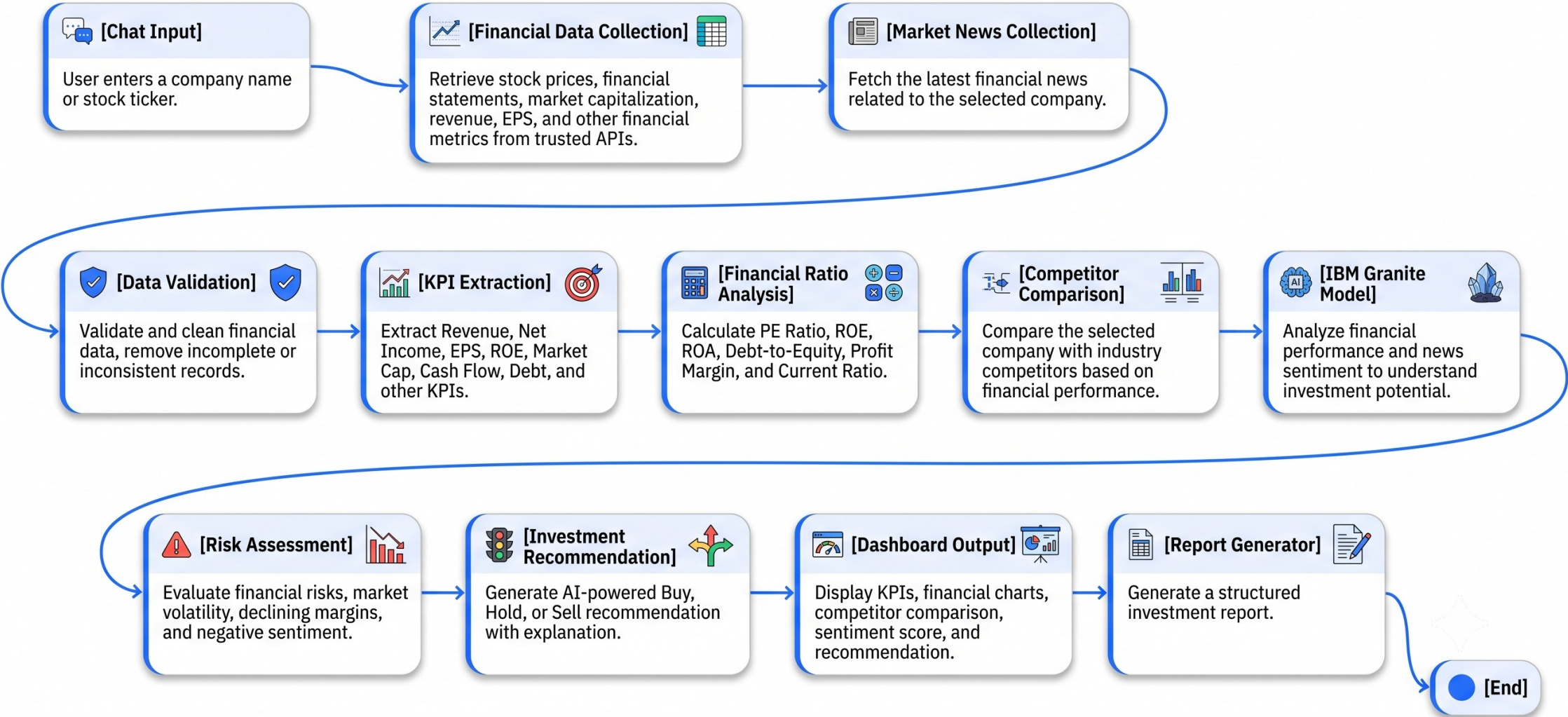
- **Langflow platform** -For designing and orchestrating multi-agent workflows visually
- **IBM Granite model** - for eg – ibm-granite-3-2-8b – For natural language understanding, reasoning, and personalization
- **IBM Cloud** – It provides the infrastructure to build, deploy, and scale the Nutrition Agent efficiently. IBM Cloud enables a **reliable, secure, and scalable environment** for deploying the Nutrition Agent.
- **RAG**– RAG to **fetch real data first, then generate intelligent answers**, making the Nutrition Agent more trustworthy and effective.
- **IBM Watsonx.ai** - For model deployment, embeddings, and AI governance
- **Vector Database** (FAISS/Chroma) – For RAG-based nutritional data retrieval

Langflow component Used -

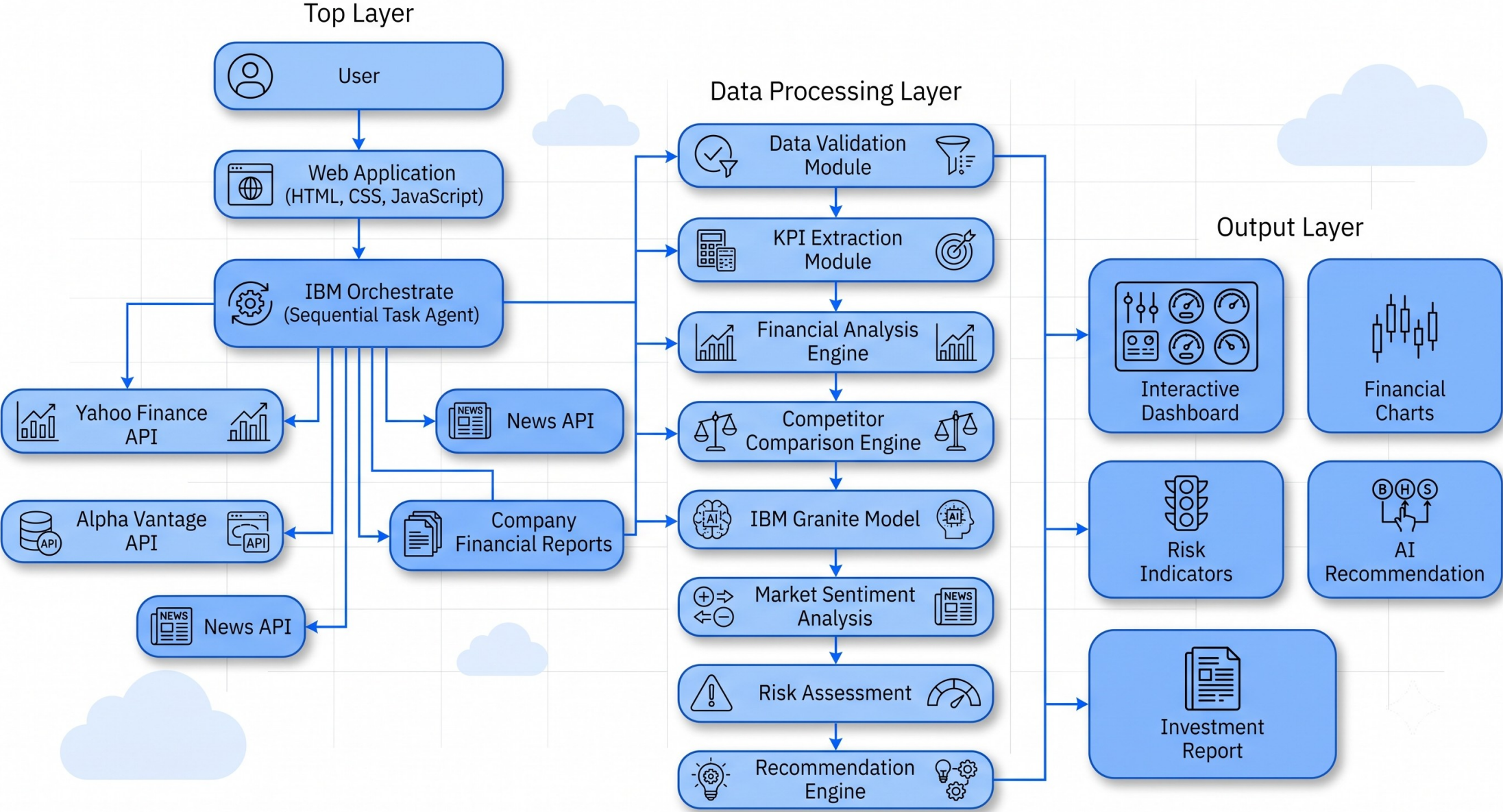


Langflow workflow -

Investment Analyst Agent Workflow



Architecture blueprint



Role of Agentic AI in the solution

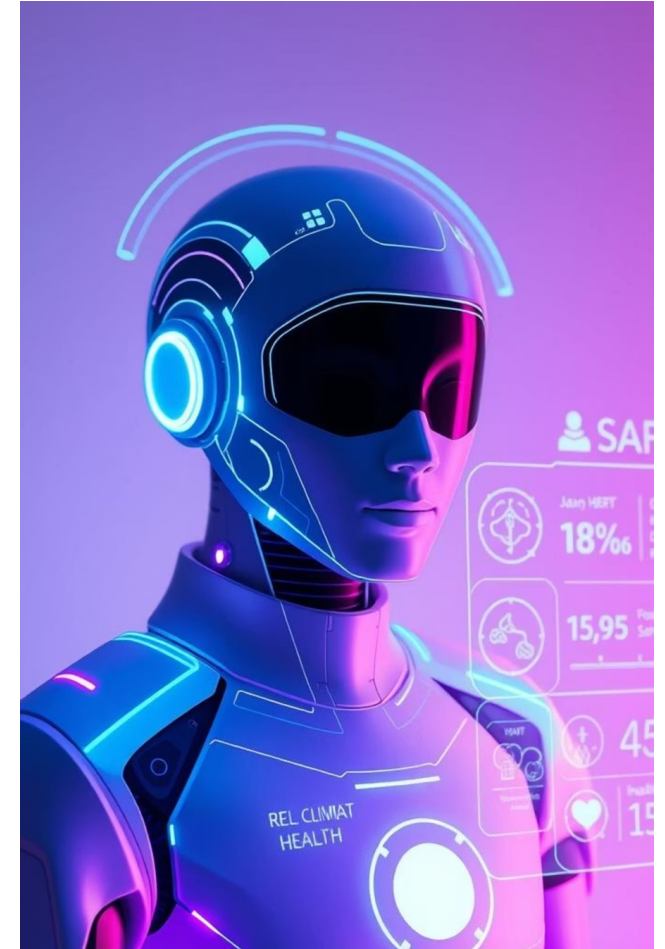
Role of Agentic AI

Agentic AI enables the Investment Analyst to function as an autonomous financial assistant instead of a traditional chatbot.

Each specialized AI agent performs a dedicated task within the investment analysis workflow.

- Data Collection Agent gathers financial information.
- Validation Agent verifies data quality.
- Financial Analysis Agent extracts KPIs and calculates ratios.
- Market Intelligence Agent analyzes financial news.
- Risk Assessment Agent detects investment risks.
- Recommendation Agent generates Buy, Hold, or Sell recommendations.

These agents collaborate to automate complex financial analysis while maintaining high accuracy and consistency.



Novelty and Uniqueness

Novelty and Uniqueness

The proposed Investment Analyst Agent stands out by combining Agentic AI, Sequential Task Automation, and IBM Granite Models into a single intelligent investment analysis platform.

Sequential Task Automation – Automates the complete investment analysis workflow from data collection to AI-powered recommendations.

Multi-Agent Intelligence – Specialized agents collaborate to gather data, analyze financial performance, assess risks, and generate insights.

Competitor Benchmarking – Compares companies using key financial ratios and performance indicators.

Interactive Dashboard – Displays KPIs, charts, risk indicators, and AI-generated Buy, Hold, or Sell recommendations.

Scalable & Enterprise-Ready – Built using IBM Orchestrate and IBM Granite Models for a reliable and intelligent investment analysis solution

Git Hub Link

Github repository

<https://github.com/sujalcodes-source/investment-analyst-agent.git>

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Future Scope

Future Enhancements

- **Integration with live stock exchanges.**
- **Portfolio management dashboard.**
- **AI-powered chatbot for financial queries.**
- **Personalized investment strategies**
- **Mutual fund recommendations.**
- **Predictive stock price forecasting.**
- **Mobile application development.**

Certificates Earned:

Credly certificate

In recognition of the commitment to achieve
professional excellence



Sujal Bhagat

Has successfully satisfied the requirements for:

Getting Started with Artificial Intelligence



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Certificates Earned:

IBM BOB Certificate

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Completion Certificate



This certificate is presented to

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Lab: Troubleshoot Your Code Using IBM Bob

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According to the Adobe Learning Manager system of record

Completion date: 14 Jul 2026 (GMT)

Learning hours: 30 mins

Thank You!

Thank you for your time and interest.